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Comorbidity Burden and Effectiveness of Immunotherapy in Metastatic Renal Cell Carcinoma

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Abstract

In this multicenter cohort study of 304 patients with metastatic renal cell carcinoma (mRCC) receiving first-line immune checkpoint inhibitor (ICI)-based combinations, a higher baseline comorbidity burden was independently associated with shorter time to treatment failure and overall survival. These findings highlight the prognostic significance of comorbidity and its relevance in individualized treatment planning for mRCC.

Background: Comorbid conditions complicate the care of patients with cancer and frequently cause exclusion of patients from clinical trials. **Methods:** Data from patients with metastatic renal cell carcinoma (mRCC) treated with immune checkpoint inhibitor (ICI)-based combinations in the first line setting were collected. The comorbidity burden was assessed at baseline by using the age-adjusted Charlson Comorbidity Index (CCI). Patients were stratified into 2 groups to predict overall survival (OS) through maximally selected rank statistics. The primary outcomes were time to treatment failure (TTF) and OS. The secondary outcome was the rate of adverse events (AEs) leading to dose reduction or treatment discontinuation. **Results:** A total of 304 patients were included. Most patients were male (73%), had clear cell RCC (91.4%), and were treated with nivolumab + ipilimumab (53.6%). The most common comorbidities were diabetes (18.4%), followed by previous myocardial infarction (12.8%), chronic kidney disease (6.6%), and chronic pulmonary disease (5.6%). After adjusting for baseline prognostic factors in mRCC including the International mRCC Database Consortium (IMDC) risk, TTF (Hazard Ratio [HR], 1.51, 95% Confidence Interval [CI], 1.09-2.10, $P = .013$) and OS (HR: 1.98, 95% CI, 1.33-2.94, $P = .001$) were worse in the CCI-high group vs. the CCI-low group. The rates of AEs leading to dose reduction or treatment discontinuation were comparable between the 2 groups. **Conclusions:** Despite similar rates of AEs leading to dose reduction or treatment discontinuation, a high comorbidity burden is associated with worse outcomes in patients with mRCC treated with first-line ICI-based therapies. Our study underscores the necessity for a multidimensional approach to assess the comorbidity burden in patients with mRCC receiving ICI-based combinations.

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Key Words: Charlson Comorbidity Index, Chronic disease, Immune checkpoint inhibitors, Kidney cancer, Renal neoplasms

Key Take-Home Messages

1. Higher baseline comorbidity burden is associated with worse outcomes in mRCC patients treated with first-line ICI-based combinations.
2. Time to treatment failure and overall survival are significantly shorter in patients with greater comorbidity.
3. Comorbidity assessment should be integrated into routine clinical decision-making to guide individualized treatment strategies.

Introduction

Immune checkpoint inhibitors (ICIs) and combinations (i.e., ICI + ICI and ICI + antivascular endothelial growth factor targeted

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therapies [anti-VEGF TT]) have launched a new era in the treatment of patients with metastatic renal cell carcinoma (mRCC).[1-5] However, numerous factors can affect the efficacy of ICIs.[6-8] In this regard, efforts to find a biomarker to select patients who will benefit more from the ICI-based treatments are ongoing.[9-12] In addition to molecular factors, clinical parameters, such as age, performance status, and body mass index, can also affect the performance of ICIs.[13,14]

Comorbidities are frequently observed in patients with cancer, often resulting from advanced age, and non–cancer-related chronic diseases.[15] In the United States, more than 40% of patients with cancer aged ≥ 65 years were found to have comorbidities,[16] with 15% having 2 or more comorbidities.[16]

Patients with multiple comorbidities may be excluded from clinical trials. This limitation can hinder the generalizability of trial results to broader patient populations.[17] Furthermore, in line with this, patients with a high comorbidity burden are less likely to receive ICIs in real-world clinical practice.[18]

There is limited data about the effect of comorbidity burden on survival outcomes, treatment response, and adverse events (AEs) profile of patients with solid tumors treated with ICIs. Most studies have shown that a higher comorbidity burden is associated with worse overall survival (OS). However, there are conflicting results regarding the impact of comorbidities on treatment failure, tumor progression, and AEs profile.[19-22]

In this study, we aimed to assess the role of comorbidity burden in patients with mRCC treated with first-line ICI-based regimens.

Methods

This study was conducted at Dana-Farber Cancer Institute and the University of Calgary. Each center obtained approval from its respective Institutional Review Board independently.

Patients and Data

Patients with mRCC receiving at least 1 dose of ICI-based regimens (ie, ICI + ICI or ICI + anti-VEGF TT) in the first line setting until a date cutoff of December 15th, 2023, were included in this study. Patients under 18 years of age were excluded. Patients' baseline demographics (eg, sex, age at starting treatment), clinical (eg, systemic treatment regimen, laboratory values, sites of metastases, nephrectomy status, Karnofsky Performance Scale (KPS), and comorbidities at the initiation of treatment), and pathologic data (eg, histological type, presence of sarcomatoid features) were collected.

Charlson Comorbidity Index (CCI)

In 1987, the CCI was initially developed to predict 1-year mortality in hospitalized patients with comorbidities and was validated in patients with breast cancer.[23] It comprises 19 comorbid conditions and uses a weighted approach for each comorbidity. Points are assigned to comorbidities based on their relative weights as follows: 1 point for previous myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, connective tissue disease, peptic ulcer disease, mild liver disease, and diabetes without chronic complications; 2 points for diabetes with chronic complications, hemiplegia or paraplegia, moderate to severe renal disease, any malignancy without metastasis, lymphoma, and leukemia; 3 points for moderate to severe liver disease; and 6 points for metastatic solid tumors and acquired immunodeficiency syndrome (AIDS). Finally, an age-adjusted model was also developed to predict 10-year mortality. In this final model, 1 additional point is assigned for each decade of age at 50 years and beyond.[23] In our study, we calculated baseline age-adjusted CCI before initiating the first-line ICI-based treatment.

Outcomes

The primary outcomes of this study were time to treatment failure (TTF) and OS. TTF was defined as the duration from the initiation of the first-line ICI-based regimen to treatment discontinuation for any reason, while OS was defined as the time from the initiation of the first-line ICI-based regimen to death. The secondary outcome was the rate of AEs leading to any dose reduction or complete ICI treatment discontinuation.

Statistical Analysis

Descriptive statistics for the dataset were presented as median values with interquartile ranges (IQR) for continuous variables and as percentages for categorical variables. Comparisons of medians between groups were conducted using the *Mann–Whitney U* test. Proportions were compared using either the *Chi-square* test or *Fisher's exact* test, with the latter applied when the sample size was too small to meet the assumptions of the *Chi-square* test. Maximally selected rank statistics were used to identify the optimal cut-off value for the age-adjusted CCI, allowing patients to be classified into "CCI-low" and "CCI-high" groups to predict overall survival.

Univariable survival analysis was performed using Kaplan–Meier curves. The log-rank test was applied to compare the survival curves. Multivariable analysis was conducted to adjust survival analysis for the following potential confounding variables: sex, presence of brain, bone, or liver metastases, International mRCC Database Consortium (IMDC) risk group, nephrectomy status, presence of sarcomatoid features, and histologic type (clear cell renal cell carcinoma (ccRCC) vs. non-ccRCC). Age and KPS were not included in the multivariable model because they were already incorporated into the age-adjusted CCI and the IMDC risk score, respectively. Patients with missing data were kept in the multivariable models by assigning them to a designated "missing" category. Cox proportional hazard regression models were utilized to calculate hazard ratios (HR) along with their corresponding 95% confidence intervals (CI). A 2-sided *P*-value of less than 0.05 was accepted as statistically significant. All analyses were performed using R (version 4.3.2) on RStudio (version 2024.12.0.467).

Results

A total of 304 patients were included in this study. The median age was 61 years (IQR: 11) in the whole cohort. Most patients were male ($n = 222$, 73%), had ccRCC ($n = 278$, 91.4%), were IMDC intermediate risk group ($n = 135$, $n = 44.4\%$), and were treated with nivolumab + ipilimumab combination ($n = 163$, 53.6%). The most common comorbidities were diabetes (18.4%), followed by previous myocardial infarction (12.8%), chronic kidney disease

Table 1 Baseline Characteristics

	All Cohort	CCI-Low	CCI-High	*P*-Value
	N = 304 (%)	*N* = 236 (%)	*N* = 68 (%)	
Age at start 1st line years, median (IQR)	63 (11)	61 (10)	72 (12)	**< .001**
Sex				.149
Female	82 (27.0%)	59 (25.0%)	23 (33.8%)	
Male	222 (73.0%)	177 (75.0%)	45 (66.2%)	
IMDC risk group				.130
Favorable	51 (16.8%)	41 (17.4%)	10 (14.7%)	
Intermediate	135 (44.4%)	108 (45.8%)	27 (39.7%)	
Poor	69 (22.7%)	47 (19.9%)	22 (32.4%)	
Clear cell RCC				.282
No	26 (8.6%)	18 (7.6%)	8 (11.8%)	
Yes	278 (91.4%)	218 (92.4%)	60 (88.2%)	
Sarcomatoid feature				.209
No	246 (80.9%)	189 (80.1%)	57 (83.8%)	
Yes	36 (11.8%)	31 (13.1%)	5 (7.4%)	
Brain, Bone, or Liver metastases				.233
No	149 (49.0%)	120 (50.8%)	29 (42.6%)	
Yes	155 (51.0%)	116 (49.2%)	39 (57.4%)	
Nephrectomy				.223
No	99 (32.6%)	81 (34.3%)	18 (26.5%)	
Yes	205 (67.4%)	155 (65.7%)	50 (73.5%)	
KPS at start 1st line				**.003**
≥ 80	225 (74.0%)	182 (77.1%)	43 (63.2%)	
< 80	49 (16.1%)	30 (12.7%)	19 (27.9%)	
1st line combination				.497
ICI + ICI	163 (53.6%)	129 (54.7%)	34 (50.0%)	
ICI + VEGF-TT	141 (46.4%)	107 (45.3%)	34 (50.0%)	

Statistically significant values are shown in bold.
Abbreviations: CCI = Charlson Comorbidity Index; ICI = immune checkpoint inhibitor; IQR = interquartile range; IMDC = international metastatic renal cell carcinoma database consortium; KPS = Karnofsky Performance Scale; RCC = renal cell carcinoma; VEGF TT = vascular endothelial growth factor targeted therapy.

(6.6%), and chronic pulmonary disease (5.6%). 11.5% (n = 35) of all patients had 2 or more comorbidities. All comorbidities are shown in Supplemental Table 1.

The optimal cut-off value for age-adjusted CCI in our study was found to be 3 points. There were 236 (77.6%) and 68 (22.4%) patients in the CCI-low (CCI≤ 3) and CCI-high (CCI> 3) groups, respectively. Baseline characteristics were comparable between the CCI-low and CCI-high groups (Table 1). However, the median age (72 vs. 61 years, P < .001) and the proportion of patients with a KPS< 80 (27.9% vs. 12.7%, P = .003) were higher in the CCI-high group compared to the CCI-low group.

The median follow-up for OS was 42.1 months (95% CI, 33.6-46.4). At the time of analysis, 136 patients (44.7%) had died, and 210 patients (69.1%) had discontinued systemic treatment with an ICI-based regimen. In univariate analysis, the median TTF (13.4 vs. 7.1 months, P = .022) and OS (46.7 vs. 29.6 months, P < .001) were longer in the CCI-low group compared to the CCI-high group (Figures 1 and 2). In multivariable analysis, the CCI-high group had a higher risk of treatment failure (HR: 1.51, 95% CI, 1.09-2.10, P = .013) and death (HR: 1.98, 95% CI, 1.33-2.94, P = .001) compared to the CCI-low group, even after adjusting for potential confounding factors (Table 2).

The proportion of AEs leading to dose reduction or treatment discontinuation was 40.1% in the whole cohort. The most common immune-related AEs (irAEs) leading to dose reduction or treatment discontinuation were hepatitis (n = 30, 9.9%), followed by colitis (n = 27, 8.9%), and rash (n = 12, 3.9%). No significant differences were observed between the CCI-low and CCI-high groups in the rates of AEs (CCI-low: 39.8% vs. CCI-high: 41.2%, P = .815), irAEs (CCI-low: 40.7% vs. CCI-high: 36.8%, P = .484), and non-irAEs (CCI-low: 19.9% vs. CCI-high: 23.5%, P = .469) that led to dose reduction or treatment discontinuation. We also did a sensitivity analysis for assessing irAEs leading to dose reduction or treatment discontinuation separately in patients receiving ICI + ICI combinations to better isolate irAEs, given the potential confounding effects in patients treated with ICI plus anti-VEGF TT. In this subgroup, the rates of irAEs were similar between the CCI-low (43.4%) and CCI-high (47.1%) groups (P = .741). The AEs profile is shown in Table 3.

Figure 1 Kaplan–Meier survival plot for time to treatment failure by CCI groups (Abbreviations: CCI = Charlson Comorbidity Index).

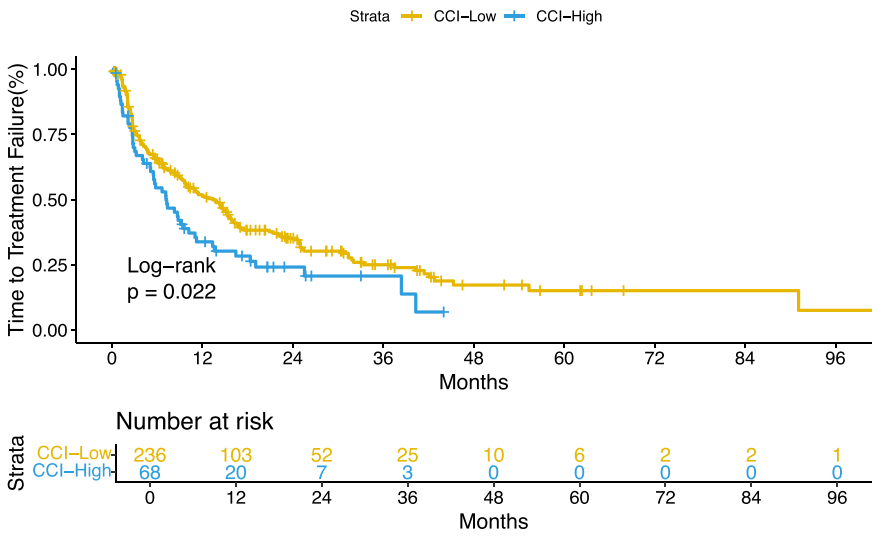
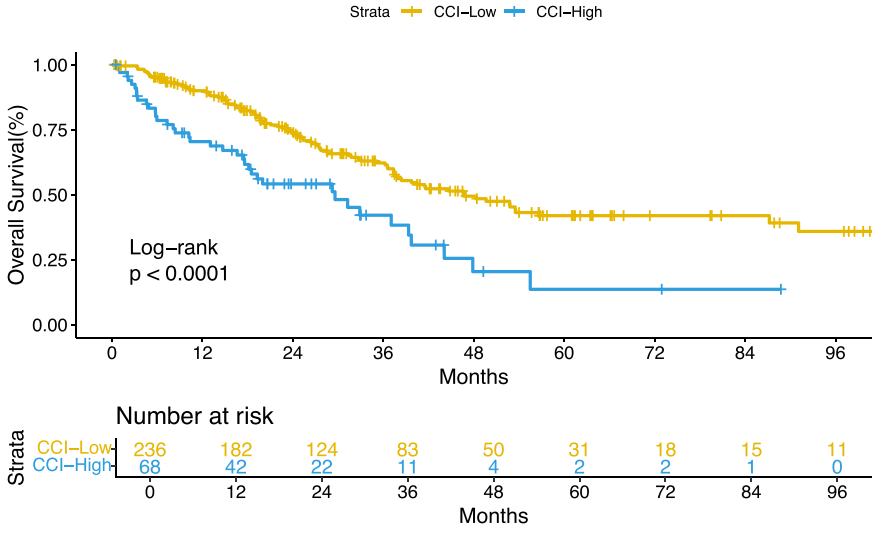


Figure 2 Kaplan–Meier survival plot for overall survival by CCI groups (Abbreviations: CCI = Charlson Comorbidity Index).



Discussion

To our knowledge, this is the first study to evaluate the impact of comorbidity burden on immunotherapy response in a cohort of patients with mRCC treated with standard ICI-based combinations. Our findings demonstrate that patients with high comorbidity burden had a higher risk of death and treatment failure, despite that the rate of AEs leading to dose reduction or treatment discontinuation was comparable between patients with high and low comorbidity burdens.

The distribution of comorbidities in our entire cohort was generally comparable to that reported in large noncancer datasets. However, the CCI-high subgroup showed a markedly higher prevalence of conditions like diabetes, myocardial infarction, and renal disease, underscoring the clinical relevance of comorbidity stratification in this population.[24,25]

Comorbid conditions might alter the immune functions in a highly complex way. A recent study evaluating the effect of preexisting comorbidities on the immune system by using single-cell

Table 2 Multivariable Cox Regression Model for TTF and OS

	TTF			OS		
	HR	95% CI	*P*-Value	HR	95% CI	*P*-Value
CCI group						
CCI-low	1.00			1.00		
CCI-high	1.51	1.09-2.10	**.013**	1.98	1.33-2.94	**.001**
Brain, Bone, or Liver metastases						
No	1.00			1.00		
Yes	1.37	1.03-1.81	**.031**	1.57	1.09-2.27	**.016**
IMDC risk group						
Favorable	1.00			1.00		
Intermediate	1.25	0.83-1.89	.281	1.22	0.73-2.03	.445
Poor	1.30	0.81-2.11	.281	1.61	0.90-2.91	.110
Sex						
Female	1.00			1.00		
Male	1.05	0.77-1.44	.757	0.65	0.44-0.94	**.023**
Nephrectomy						
No	1.00			1.00		
Yes	0.89	0.64-1.24	.488	0.73	0.48-1.12	.152
Sarcomatoid feature						
No	1.00			1.00		
Yes	0.97	0.61-1.55	.904	1.22	0.69-2.15	.501
Clear cell RCC						
No	1.00			1.00		
Yes	0.79	0.48-1.28	.332	0.36	0.21-0.61	**< .001**

Statistically significant values are shown in bold.
Abbreviations: CCI = Charlson Comorbidity Index; CI = confidence interval; HR = hazard ratio; IMDC = international metastatic renal cell carcinoma database consortium; OS = overall survival; RCC = renal cell carcinoma; TTF = time to treatment failure.

mapping of peripheral blood concluded that different comorbidities could contribute to immune dysregulation by affecting distinct immune pathways.[26] Metabolic syndrome could impair both innate and adaptive immunity, while chronic kidney disease could disrupt the regulation of lymphoid and myeloid immune functions.[26] Consequently, patients with multiple comorbidities may have worse immune system function, leading to therapy resistance and cancer progression.

There is a dynamic interplay between comorbidities and the gut microbiome.[27,28] Obesity and diabetes are well-known comorbidities leading to an imbalance of the gut microbiome, also known as dysbiosis.[27-29] Given that dysbiosis contributes to cancer development and progression, comorbidities may also promote anticancer treatment resistance through this indirect pathway.[30-32]

The relationship between comorbidities and the immune system plays a crucial role regarding the comorbidity-driven cancer progression and treatment resistance. However, comorbidities can also promote tumor growth and treatment resistance through alternative mechanisms. In this regard, a study showed that renin-angiotensin-aldosterone pathway signaling molecules released in the context of heart failure were independent risk factors for tumor growth.[33] Chronic hyperglycemia could trigger protein kinase C and NF-κB, thus promoting oncogenesis, angiogenesis, and metastasis.[34] Furthermore, clinical conditions linked to comorbidities, such as polypharmacy, end-organ dysfunction, and poor performance status, can also exacerbate cancer progression.[35-37]

Studies examining the relationship between comorbidities and cancer immunotherapy revealed that patients with a high comorbid disease burden had worse OS compared to those with a low or no comorbid burden.[19-22] Increased mortality is inherently expected in cancer patients with comorbidities, but the impact of comorbidities on systemic therapy efficacy and toxicity is poorly researched.[19-21] In 1 study, patients with head and neck cancer treated with ICIs with low comorbidity burden, had an improved OS and PFS alongside increased risk for irAEs compared to those with high comorbidity burden.[20] In contrast to these findings regarding irAEs, a higher CCI score was found to be associated with increased rates of irAEs in a study including patients with various cancer types treated with ICIs.[38] This study also concluded that various malignancies may have unique risk profiles of irAEs. For instance, the risk of irAEs was higher in patients with lung cancer or RCC who had a higher CCI score, while no relation was found between the CCI score and risk of irAEs in patients with melanoma.[38] As previously noted, each comorbidity uniquely influences the immune system function, and additional factors (eg, polypharmacy, end-organ dysfunction) linked to these comorbidities may further contribute to the variability in the study outcomes.[26,35-37] Variability in findings across cancer types may be driven by the intricate relationship between tumor biology, treatment approach, and the nature of underlying comorbidities. Unlike lung or head and neck cancers, which are often treated with chemotherapy or single-agent immunotherapy with distinct toxicity considerations, mRCC

Table 3 Adverse Events Leading to Dose Reduction or Treatment Discontinuation

	All Cohort	CCI-Low	CCI-High
All AEs	**122 (40.1%)**	**94 (39.8%)**	**28 (41.2%)**
irAEs	**121 (39.8%)**	**96 (40.7%)**	**25 (36.8%)**
Hepatitis	30 (9.9%)	28 (9.2%)	2 (0.7%)
Colitis	27 (8.9%)	20 (6.6%)	7 (2.3%)
Rash	12 (3.9%)	8 (2.6%)	4 (1.3%)
Arthritis	9 (3.0%)	9 (3.0%)	0 (0.0%)
Pneumonitis	9 (3.0%)	6 (2.0%)	3 (1.0%)
Adrenalitis	8 (2.6%)	7 (2.3%)	1 (0.3%)
Pancreatitis	6 (2.0%)	5 (1.6%)	1 (0.3%)
Thyroiditis	5 (1.6%)	4 (1.3%)	1 (0.3%)
Infusion	3 (1.0%)	1 (0.3%)	2 (0.7%)
Nephritis	3 (1.0%)	2 (0.7%)	1 (0.3%)
Autoimmune cytopenia	2 (0.7%)	1 (0.3%)	1 (0.3%)
Hypophysitis	2 (0.7%)	1 (0.3%)	1 (0.3%)
Myocarditis	2 (0.7%)	2 (0.7%)	0 (0.0%)
Pericarditis	2 (0.7%)	2 (0.7%)	0 (0.0%)
Neurologic	2 (0.7%)	1 (0.3%)	1 (0.3%)
Blepharitis	1 (0.3%)	1 (0.3%)	0 (0.0%)
Encephalitis	1 (0.3%)	1 (0.3%)	0 (0.0%)
Unknown	1 (0.3%)	1 (0.3%)	0 (0.0%)
Non-irAEs	**63 (20.7%)**	**47 (19.9%)**	**16 (23.5%)**
Fatigue	14 (4.6%)	11 (3.6%)	3 (1.0%)
Gastrointestinal	14 (4.6%)	11 (3.6%)	3 (1.0%)
Hypertension	11 (3.6%)	7 (2.3%)	4 (1.3%)
Mucositis	9 (3.0%)	5 (1.6%)	4 (1.3%)
Hand-Foot syndrome	8 (2.6%)	8 (2.6%)	0 (0.0%)
Anorexia	6 (2.0%)	6 (2.0%)	0 (0.0%)
Bleeding	3 (1.0%)	3 (1.0%)	0 (0.0%)
Stroke	3 (1.0%)	1 (0.3%)	2 (0.7%)
Arrhythmia	2 (0.7%)	1 (0.3%)	1 (0.3%)
Dizziness	2 (0.7%)	1 (0.3%)	1 (0.3%)
Liver	2 (0.7%)	1 (0.3%)	1 (0.3%)
Acute Coronary syndrome	2 (0.7%)	2 (0.7%)	0 (0.0%)
Electrolyte imbalance	1 (0.3%)	1 (0.3%)	0 (0.0%)
Myalgia	1 (0.3%)	0 (0.0%)	1 (0.3%)
Proteinuria	1 (0.3%)	1 (0.3%)	0 (0.0%)

Groups in bold represent main categories of AEs.
Abbreviations: AEs = adverse events; CCI = Charlson Comorbidity Index; irAEs = immune related adverse events.

is commonly managed with ICI-based combinations along with anti-VEGF TT, which may have a different tolerability profile in patients with existing health conditions. In our cohort, the absence of a difference in irAE by CCI status may reflect RCC-specific immunologic responses or differences in clinical practice regarding toxicity documentation. These findings highlight the importance of studying comorbidity effects in the context of individual cancer types, rather than extrapolating conclusions across diverse disease settings.

Besides the negative impact of increased comorbidity burden on immunotherapy, comorbid conditions may also decrease the efficacy of anti-VEGF TT in patients with mRCC. A real-world study conducted in the United Kingdom revealed that a high comorbidity burden was associated with worse OS and PFS in patients with mRCC receiving cabozantinib.[39]

We also showed that female patients had higher risk for death in the multivariable analysis. A large meta-analysis comparing female and male patients with cancer treated with immunotherapy revealed that men had better survival compared to women.[40] Sex hormones and chromosomes might affect the tumor microenvironment and response to immunotherapy.[41]

Our study has several limitations. First, it is essential to take into account the retrospective nature of the study. Therefore, the missing data from 49 patients may have limited the ability to detect a statistically significant difference in the multivariable analysis of OS and TTF among the IMDC risk groups. Second, we did not

collect data on comorbidities during and after treatment. Therefore, since we evaluated comorbidities documented before initiating first-line ICI-based regimens, additional comorbidities could have influenced survival outcomes. Third, we only assessed irAEs leading to dose reduction or treatment discontinuation. The relatively low frequency of AEs resulting in dose reduction or treatment discontinuation may be attributed to the retrospective design of the study, which is inherently subject to incomplete or variable documentation of toxicities in clinical practice. Fourth, we did not assess the individual impact of specific comorbidities on outcomes due to the limited representation of patients with certain conditions, such as AIDS, dementia, hemiplegia, and liver disease in our cohort. Fifth, cause of death data were not uniformly available in participating centers, limiting our ability to distinguish RCC-related from non–RCC-related mortality by CCI group. Nonetheless, the observed survival differences remained clinically meaningful and highlighted the prognostic impact of comorbidity burden in this group of patients. Finally, while we categorized comorbidity burden into 2 groups to preserve statistical power, we acknowledge that distinguishing patients with no comorbidities from those with mild comorbidity (eg, CCI 1-3) may offer additional insights and should be explored in future larger studies.

Conclusion

This was the first study to assess the effect of comorbidity burden on the efficacy and AEs profile in patients with mRCC treated with ICI-based combinations. We showed that a high comorbidity burden was an independent risk factor for death and treatment failure. However, the rates of AEs leading to dose reduction or treatment discontinuation were similar in high and low comorbidity burden groups. The findings of this study suggest that incorporating comorbidity evaluation into the treatment decision-making process could benefit patients with mRCC receiving first-line ICI-based regimens. Additionally, this study emphasizes the importance of further research to explore the interactions between the immune system and comorbidities in mRCC, which may provide valuable insights for optimizing treatment strategies.

Clinical Practice Points

Immune checkpoint inhibitor (ICI)-based combinations are widely used as first-line treatment for metastatic renal cell carcinoma (mRCC), yet the prognostic impact of comorbidity burden has not been well defined in real-world settings. In a multicenter cohort of 304 patients, we found that higher baseline comorbidity burden was independently associated with shorter time to treatment failure and overall survival. These findings suggest that comorbidity assessment should be routinely integrated into clinical decision-making to better individualize treatment strategies and improve prognostication in patients with mRCC.

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CRediT authorship contribution statement

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Supplementary material

Supplemental Table 1 Comorbidities

	All Cohort	CCI-Low	CCI-High
	N = 304 (%)	*N* = 236 (%)	*N* = 68 (%)
Diabetes	56 (18.4%)	22 (9.3%)	34 (50.0%)
Previous myocardial infarction	39 (12.8%)	16 (6.8%)	23 (33.8%)
Renal disease	20 (6.6%)	6 (2.5%)	14 (20.6%)
Chronic pulmonary disease	17 (5.6%)	6 (2.5%)	11 (16.2%)
Congestive heart failure	16 (5.3%)	4 (1.7%)	12 (17.6%)
Peripheral vascular disease	10 (3.3%)	3 (1.3%)	7 (10.3%)
Cerebrovascular disease	10 (3.3%)	1 (0.4%)	9 (13.2%)
Rheumatologic disease	8 (2.6%)	3 (1.3%)	5 (7.4%)
Peptic ulcer disease	4 (1.3%)	0 (0.0%)	4 (5.9%)
Dementia	2 (0.7%)	0 (0.0%)	2 (2.9%)
Hemiplegia	2 (0.7%)	0 (0.0%)	2 (2.9%)
Liver disease	1 (0.3%)	0 (0.0%)	1 (1.5%)
Lymphoma	1 (0.3%)	0 (0.0%)	1 (1.5%)